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AI



OpenAI and Google
outdo the mathletes, but
not each other

AI models from [OpenAI](#) and [Google DeepMind](#) achieved gold-medal scores in the 2025 International Math Olympiad (IMO), one of the world's oldest and most challenging high school-level math competitions, the companies independently announced in recent days.

The results underscore just how fast AI systems are advancing, and yet, how evenly matched Google and OpenAI seem to be in the AI race. AI companies are competing fiercely for the public perception of being ahead in the AI race: an intangible battle of “vibes” that can have big implications for securing top AI talent. A lot of AI researchers come from backgrounds in competitive math, so benchmarks like IMO mean more than others.

Last year, [Google scored a silver medal at IMO](#) using a “formal” system, meaning it required humans to translate problems into a machine-readable format. This year, both OpenAI and Google entered “informal” systems into the competition, which were able to ingest questions and generate proof-based answers in natural language. Both companies claim their AI models correctly answered five out of six questions on IMO's test, scoring higher than most high school students and Google's AI model from last year, without requiring any human-machine translation.

In interviews with TechCrunch, researchers behind OpenAI and Google's IMO efforts claimed that

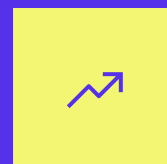
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these gold-medal performances represent breakthroughs around AI reasoning models in non-verifiable domains. While AI reasoning models tend to do well on questions with straightforward answers, such as simple math or coding tasks, these systems struggle on tasks with more ambiguous solutions, such as buying a great chair or helping with complex research.

However, Google is raising questions around how OpenAI conducted and announced its gold-medal IMO performance. After all, if you're going to enter AI models into a math contest for high schoolers, you might as well argue like teenagers.

Shortly after OpenAI announced its feat on Saturday morning, [Google DeepMind's CEO](#) and researchers took to social media to [slam OpenAI for announcing its gold medal prematurely](#) — shortly after IMO announced which high schoolers had won the competition on Friday night — and for not having their model's test officially evaluated by IMO.

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Demis Hassabis  · Jul 21, 2025



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We achieved this year's impressive result using an advanced version of Gemini Deep Think (an enhanced reasoning mode for complex problems). Our model operated end-to-end in natural language, producing rigorous mathematical proofs directly from the official problem descriptions –



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Btw as an aside, we didn't announce on Friday because we respected the IMO Board's original request that all AI labs share their results only after the official results had been verified by independent experts & the students had rightly received the acclamation they deserved

4:47 PM · Jul 21, 2025



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Thang Luong, a Google DeepMind senior researcher and lead for the IMO project, told TechCrunch that Google waited to announce its IMO results to respect the students participating in the competition.



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Luong said that Google has been working with IMO's organizers since last year in preparation for the test and wanted to have the IMO president's blessing and official grading before [announcing its official results, which it did on Monday morning](#).

"The IMO organizers have their grading guideline," Luong said. "So any evaluation that's not based on that guideline could not make any claim about gold-medal level [performance]."

Noam Brown, a senior OpenAI researcher who worked on the IMO model, told TechCrunch that IMO reached out to OpenAI a few months ago about participating in a formal math competition, but the ChatGPT-maker declined because it was working on natural language systems that it thought were more worth pursuing. Brown says OpenAI didn't know IMO was conducting an informal test with Google.

OpenAI says it hired third-party evaluators — three former IMO medalists who understood the grading

system — to grade its AI model’s performance. After OpenAI learned of its gold-medal score, Brown said the company reached out to IMO, which then told the company to wait to announce until after IMO’s Friday night award ceremony.

IMO did not respond to TechCrunch’s request for comment.

Google isn’t necessarily wrong here — it did go through a more official, rigorous process to achieve its gold-medal score — but the debate may miss the bigger picture: AI models from several leading AI labs are improving quickly. Countries from around the world sent their brightest students to compete at IMO this year, and just a few percent of them scored as well as OpenAI and Google’s AI models did.

While OpenAI used to have a significant lead over the industry, it certainly feels as though the race is more closely matched than any company would like to admit. OpenAI is expected to release GPT-5 in the coming months, and the company certainly hopes to give off the impression that it still leads the AI industry.

Topics: [AI](#) [Competitive math](#) [Google](#) [OpenAI](#)

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Maxwell Zeff

Senior AI Reporter | [X](#)

Maxwell Zeff is a senior reporter at TechCrunch specializing in AI. Previously with Gizmodo, Bloomberg,...

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IMAGE CREDITS: GOOGLE DEEPMIND

AI



Google rolls out Gemini Deep Think AI, a reasoning model that tests multiple ideas in parallel

Maxwell Zeff — 4:00 AM PDT · August 1, 2025

Google DeepMind is rolling out [Gemini 2.5 Deep Think](#), which, the company says, is its most advanced AI reasoning model, able to answer questions by exploring and considering multiple ideas simultaneously and then using those outputs to choose the best answer.

Subscribers to Google's \$250-per-month [Ultra](#) subscription will gain access to Gemini 2.5 Deep Think in the Gemini app starting Friday.

First unveiled in May at Google I/O 2025, Gemini 2.5 Deep Think is Google's first publicly available multi-agent model. These systems spawn multiple AI

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agents to tackle a question in parallel, a process that uses significantly more computational resources than a single agent, but tends to result in better answers.

Google used a variation of Gemini 2.5 Deep Think to [score a gold medal](#) at this year’s International Math Olympiad (IMO).

Alongside Gemini 2.5 Deep Think, the company says it is releasing the model it used at the IMO to a select group of mathematicians and academics. Google says this AI model “takes hours to reason,” instead of seconds or minutes like most consumer-facing AI models. The company hopes the IMO model will enhance research efforts, and aims to get feedback on how to improve the multi-agent system for academic use cases.

Google notes that the Gemini 2.5 Deep Think model is a significant improvement over what it announced at I/O. The company also claims to have developed “novel reinforcement learning techniques” to encourage Gemini 2.5 Deep Think to make better use of its reasoning paths.

“Deep Think can help people tackle problems that require creativity, strategic planning and making improvements step-by-step,” said Google in a [blog post](#) shared with TechCrunch.



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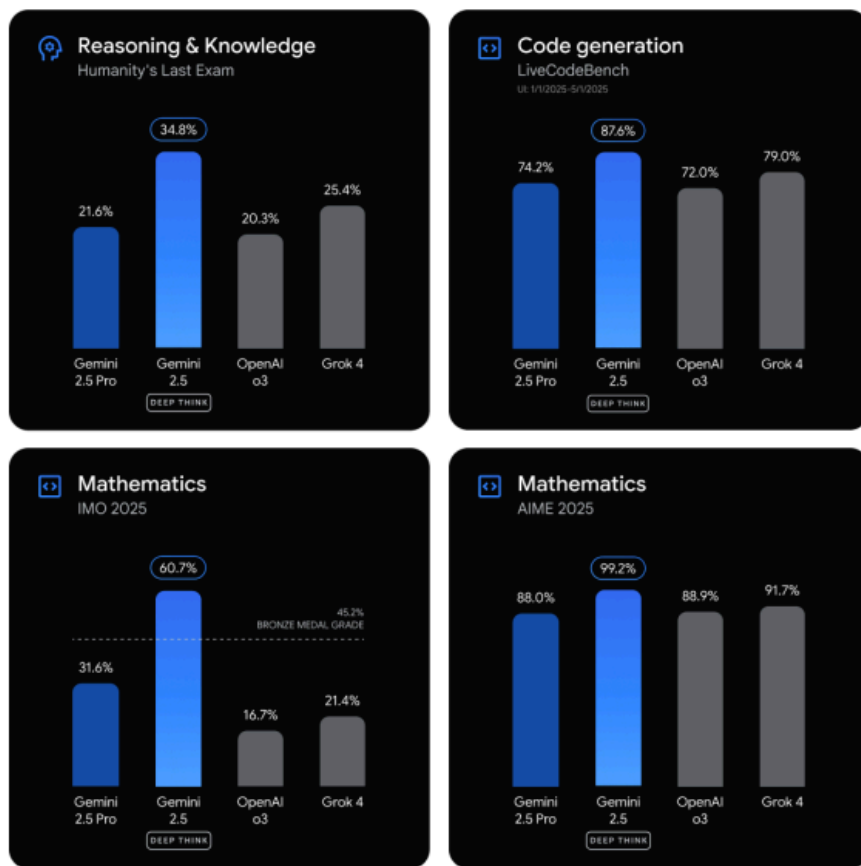
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The company says Gemini 2.5 Deep Think achieves state-of-the-art performance on Humanity’s Last Exam (HLE) — a challenging test measuring AI’s ability to answer thousands of crowdsourced questions across math, humanities, and science. Google claims its model scored 34.8% on HLE (without tools), compared to xAI’s Grok 4, which scored 25.4%, and OpenAI’s o3, which scored 20.3%.

Google also says Gemini 2.5 Deep Think outperforms AI models from OpenAI, xAI, and Anthropic on LiveCodeBench 6, a challenging test of competitive coding tasks. Google’s model scored 87.6%, whereas Grok 4 scored 79%, and OpenAI’s o3 scored 72%.

Comparison across models without tools enabled



BENCHMARK SCORES

IMAGE CREDITS:GOOGLE

Gemini 2.5 Deep Think automatically works with tools such as code execution and Google Search, and the company says it's capable of producing "much longer responses" than traditional AI models.

In Google's testing, the model produced more detailed and aesthetically pleasing web development tasks compared to other AI models. The company claims the model could aid researchers and "potentially accelerate the path to discovery."

Design and create a very creative, elaborate, and detailed voxel art scene of a pagoda in a beautiful garden with trees, including some cherry blossoms. Make the scene impressive and varied and use colorful voxels. Use whatever libraries to get this done but make sure I can paste it all into a single HTML file and open it in Chrome.



Gemini 2.5 Flash



Gemini 2.5 Pro



Gemini 2.5 Deep Think

ART SCENES MADE BY GOOGLE'S AI

IMAGE CREDITS:GOOGLE

It seems that several leading AI labs are converging around the multi-agent approach.

Elon Musk's xAI recently released a multi-agent system of its own, [Grok 4 Heavy](#), which it says was able to achieve industry-leading performance on several benchmarks. OpenAI researcher Noam Brown said on a [podcast](#) that the unreleased AI model the company used to achieve a gold medal at this year's International Math Olympiad was also a multi-agent system. Meanwhile, [Anthropic's Research agent](#), which generates thorough research briefs, is also powered by a multi-agent system.

Despite the strong performance, it seems that multi-agent systems are even costlier to serve than traditional AI models. That means tech companies may keep these systems gated behind their most expensive subscription plans, which xAI and now Google have chosen to do.

In the coming weeks, Google says it plans to share Gemini 2.5 Deep Think with a select group of testers via the Gemini API. The company says it wants to better understand how developers and enterprises may use its multi-agent system.

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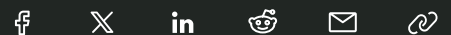
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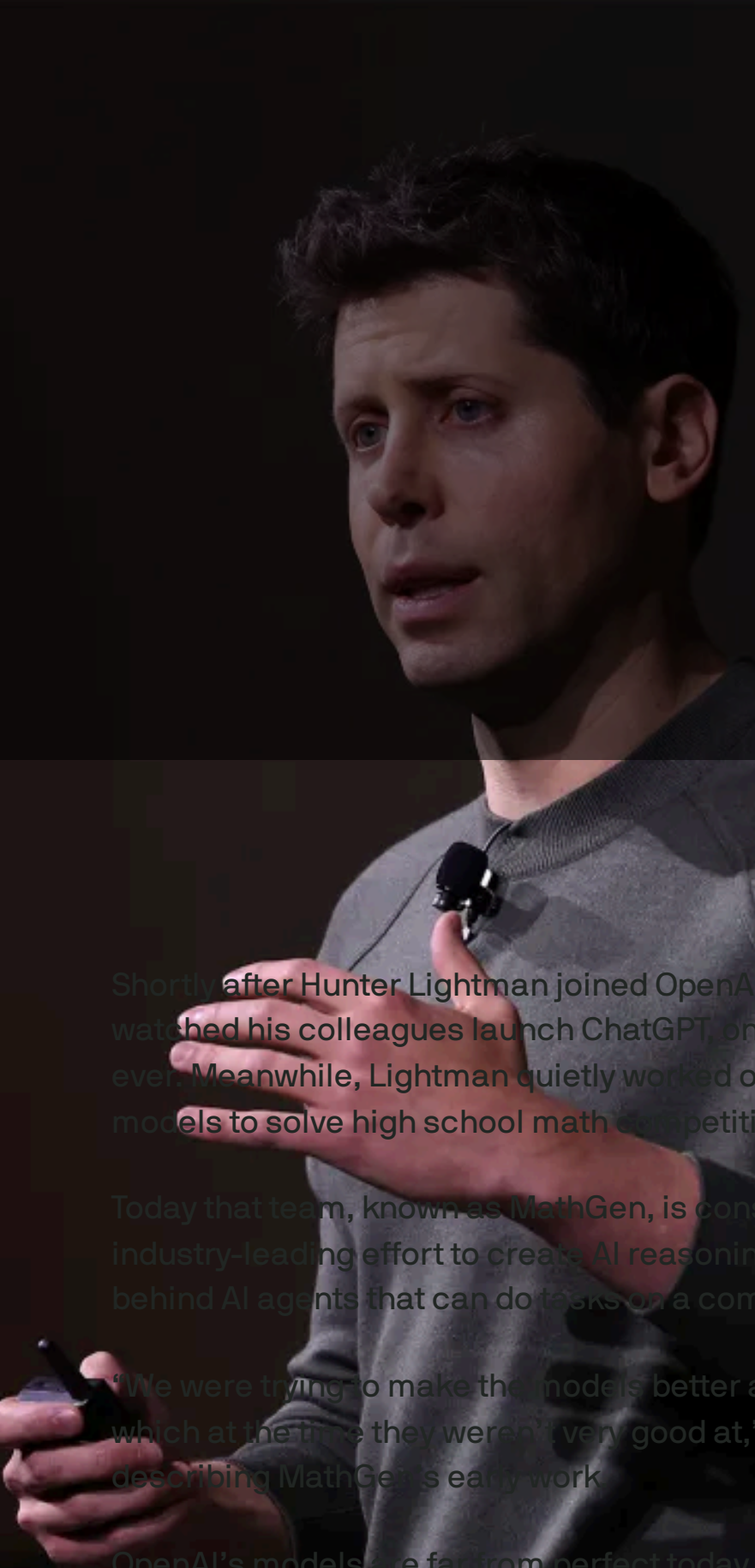
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Inside OpenAI's quest to make AI do anything for you



How OpenAI's first reasoning model sparked Silicon Valley's obsession with general-purpose AI agents — and what comes next.

Maxwell Zeff

7:00 AM PDT · August 3, 2025

Shortly after Hunter Lightman joined OpenAI as a researcher in 2022, he watched his colleagues launch ChatGPT, one of the fastest-growing products ever. Meanwhile, Lightman quietly worked on a team teaching OpenAI's models to solve high school math competitions.

Today that team, known as MathGen, is considered instrumental to OpenAI's industry-leading effort to create AI reasoning models: the core technology behind AI agents that can do tasks on a computer like a human would.

"We were trying to make the models better at mathematical reasoning, which at the time they weren't very good at," Lightman told TechCrunch, describing MathGen's early work.

OpenAI's models are far from perfect today — the company's latest AI systems still hallucinate and its agents struggle with complex tasks.

IMAGE CREDITS: JUSTIN SULLIVAN / GETTY IMAGES

But its state-of-the-art models have improved significantly on mathematical reasoning. One of OpenAI's models recently won a [gold medal](#) at the International Mathematical Olympiad (IMO), a math competition for the world's brightest high school students. OpenAI believes these reasoning capabilities will translate to other subjects, and ultimately power general-purpose agents that the company has always dreamed of building.

ChatGPT was a happy accident — a low-key research preview turned viral consumer business — but OpenAI's agents are the product of a yearslong, deliberate effort within the company.

“Eventually, you'll just ask the computer for what you need and it'll do all of these tasks for you,” said OpenAI CEO Sam Altman at the company's [first developer conference](#) in 2023. “These capabilities are often talked about in the AI field as agents. The upsides of this are going to be tremendous.”



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OPENAI CEO SAM ALTMAN SPEAKS DURING THE OPENAI DEVDAY EVENT ON NOVEMBER 06, 2023, IN SAN FRANCISCO, CALIFORNIA.

IMAGE CREDITS: JUSTIN SULLIVAN / GETTY IMAGES

Whether agents will meet Altman's vision remains to be seen, but OpenAI shocked the world with the release of [its first AI reasoning model, o1](#), in the fall of 2024. Less than a year later, the 21 foundational researchers behind that breakthrough are the most highly sought-after talent in Silicon Valley.

Mark Zuckerberg [recruited](#) five of the o1 researchers to work on Meta's new superintelligence-focused unit, offering some compensation packages north of \$100 million. One of them, Shengjia Zhao, was recently [named](#) chief scientist of Meta Superintelligence Labs.

The reinforcement learning renaissance

The rise of OpenAI's reasoning models and agents are tied to a machine learning training technique known as reinforcement learning (RL). RL provides feedback to an AI model on whether its choices were correct in simulated environments.

RL has been used for decades. For instance, in 2016, about a year after OpenAI was founded in 2015, an AI system created by Google DeepMind using RL, [AlphaGo](#), gained global attention after beating a world champion in the board game Go.

SOUTH KOREAN PROFESSIONAL GO PLAYER LEE SE-DOL (R) PREPARES FOR HIS FOURTH MATCH AGAINST GOOGLE'S ARTIFICIAL INTELLIGENCE PROGRAM, ALPHAGO, DURING THE GOOGLE DEEPMIND CHALLENGE MATCH ON MARCH 13, 2016, IN SEOUL, SOUTH KOREA.

IMAGE CREDITS:GOOGLE / GETTY IMAGES

Around that time, one of OpenAI's first employees, Andrej Karpathy, began pondering how to leverage RL to create an AI agent that could use a computer. But it would take years for OpenAI to develop the necessary models and training techniques.

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By 2018, OpenAI pioneered its first large language model in the GPT series, pretrained on massive amounts of internet data and large clusters of GPUs. GPT models excelled at text processing, eventually leading to ChatGPT, but struggled with basic math.

It took until 2023 for OpenAI to achieve a breakthrough, initially dubbed "Q*" and then "Strawberry," by combining LLMs, RL, and a technique called "test-

time computation.” The latter gave the models extra time and computing power to plan and work through problems, verifying its steps, before providing an answer.

This allowed OpenAI to introduce a new approach called “chain-of-thought” (CoT), which improved AI’s performance on math questions the models hadn’t seen before.

“I could see the model starting to reason,” said OpenAI researcher Ahmed El-Kishky. “It would notice mistakes and backtrack, it would get frustrated. It really felt like reading the thoughts of a person.”

Though individually these techniques weren’t novel, OpenAI uniquely combined them to create Strawberry, which directly led to the development of o1. OpenAI quickly identified that the planning and fact-checking abilities of AI reasoning models could be useful to power AI agents.

“We had solved a problem that I had been banging my head against for a couple of years,” said Lightman. “It was one of the most exciting moments of my research career.”

Scaling reasoning

With AI reasoning models, OpenAI determined it had two new axes that would allow it to improve AI models: using more computational power during the post-training of AI models, and giving AI models more time and processing power while answering a question.

“OpenAI, as a company, thinks a lot about not just the way things are, but the way things are going to scale,” said Lightman.

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make AI systems capable of completing complex tasks.

Eventually, the work of Selsam's Agents team became part of a larger project to develop the o1 reasoning model, with leaders including OpenAI co-founder Ilya Sutskever, chief research officer Mark Chen, and chief scientist Jakub Pachocki.

ILYA SUTSKEVER, COMPUTER SCIENTIST AND CO-FOUNDER AND FORMER CHIEF SCIENTIST OF OPENAI.
IMAGE CREDITS: JACK GUEZ / AFP / GETTY IMAGES

OpenAI would have to divert precious resources — mainly talent and GPUs — to create o1. Throughout OpenAI's history, researchers have had to negotiate with company leaders to obtain resources; demonstrating breakthroughs was a surefire way to secure them.

"One of the core components of OpenAI is that everything in research is bottom up," said Lightman. "When we showed the evidence [for o1], the company was like, 'This makes sense, let's push on it.'"

Some former employees say that the startup's mission to develop AGI was the key factor in achieving breakthroughs around AI reasoning models. By

focusing on developing the smartest-possible AI models, rather than products, OpenAI was able to prioritize o1 above other efforts. That type of large investment in ideas wasn't always possible at competing AI labs.

The decision to try new training methods proved prescient. By late 2024, several leading AI labs started seeing [diminishing returns](#) on models created through traditional pretraining scaling. Today, much of the AI field's momentum comes from advances in reasoning models.

What does it mean for an AI to “reason”?

In many ways, the goal of AI research is to re-create human intelligence with computers. Since the launch of o1, ChatGPT's UX has been filled with more human-sounding features such as “thinking” and “reasoning.”

When asked whether OpenAI's models were truly reasoning, El-Kishky hedged, saying he thinks about the concept in terms of computer science.

“We're teaching the model how to efficiently expend compute to get an answer. So if you define it that way, yes, it is reasoning,” said El-Kishky.

Lightman takes the approach of focusing on the model's results and not as much on the means or their relation to human brains.



“If the model is doing hard things, then it is doing whatever necessary approximation of reasoning it needs in order to do that,” said Lightman. “We can call it reasoning, because it looks like these reasoning traces, but it’s all just a proxy for trying to make AI tools that are really powerful and useful to a lot of people.”

OpenAI’s researchers note people may disagree with their nomenclature or definitions of reasoning — and surely, [critics have emerged](#) — but they argue it’s less important than the capabilities of their models. Other AI researchers tend to agree.

Nathan Lambert, an AI researcher with the non-profit AI2, compares AI reasoning modes to airplanes in a [blog post](#). Both, he says, are man-made systems inspired by nature — human reasoning and bird flight, respectively — but they operate through entirely different mechanisms. That doesn’t make them any less useful, or any less capable of achieving similar outcomes.

A group of AI researchers from OpenAI, Anthropic, and Google DeepMind agreed in a recent [position paper](#) that AI reasoning models are not well understood today, and more research is needed. It may be too early to confidently claim what exactly is going on inside them.

The next frontier: AI agents for subjective tasks

The AI agents on the market today work best for well-defined, verifiable domains such as coding. OpenAI’s [Codex agent](#) aims to help software engineers offload simple coding tasks. Meanwhile, Anthropic’s models have become particularly [popular](#) in AI coding tools like Cursor and Claude Code — these are some of the first AI agents that people are willing to [pay up for](#).

However, general-purpose AI agents like OpenAI’s [ChatGPT agent](#) and Perplexity’s [Comet](#) struggle with many of the complex, subjective tasks people want to automate. When trying to use these tools for online shopping or finding a long-term parking spot, I’ve found the agents take longer than I’d like and make [silly mistakes](#).

Agents are, of course, early systems that will undoubtedly improve. But researchers must first figure out how to better train the underlying models to complete tasks that are more subjective.



IMAGE CREDITS: JONATHAN RAA/NURPHOTO / GETTY IMAGES

“Like many problems in machine learning, it’s a data problem,” said Lightman when asked about the limitations of agents on subjective tasks. “Some of the research I’m really excited about right now is figuring out how to train on less verifiable tasks. We have some leads on how to do these things.”

Noam Brown, an OpenAI researcher who helped create the IMO model and o1, told TechCrunch that OpenAI has new general-purpose RL techniques that allow them to teach AI models skills that aren’t easily verified. This was how the company built the model that achieved a gold medal at IMO, he said.

OpenAI’s IMO model was a newer AI system that spawns multiple agents, which then simultaneously explore several ideas and then choose the best possible answer. These types of AI models are becoming more popular; [Google](#) and [xAI](#) have recently released state-of-the-art models using this technique.

“I think these models will become more capable at math, and I think they’ll get more capable in other reasoning areas as well,” said Brown. “The progress has been incredibly fast. I don’t see any reason to think it will slow down.”

These techniques may help OpenAI's models become more performant, gains that could show up in the company's upcoming GPT-5 model. OpenAI hopes to assert its dominance over competitors with the launch of GPT-5, ideally offering the [best AI model](#) to power agents for developers and consumers.

But the company also wants to make its products simpler to use. El-Kishky says OpenAI wants to develop AI agents that intuitively understand what users want, without requiring them to select specific settings. He says OpenAI aims to build AI systems that understand when to call up certain tools, and how long to reason for.

These ideas paint a picture of an ultimate version of ChatGPT: an agent that can do anything on the internet for you, and understand how you want it to be done. That's a much different product than what ChatGPT is today, but the company's research is squarely headed in this direction.

While OpenAI undoubtedly led the AI industry a few years ago, the company now faces a tranche of worthy opponents. The question is no longer just whether OpenAI can deliver its agentic future, but can the company do so before Google, Anthropic, xAI, or Meta beat them to it?

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Maxwell Zeff

Senior AI Reporter | [X](#)

Maxwell Zeff is a senior reporter at TechCrunch specializing in AI. Previously with Gizmodo, Bloomberg, and MSNBC, Zeff has covered the rise of AI and the Silicon Valley Bank crisis. He is based in San Francisco. Wh...

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Harmonic, the Robinhood CEO's AI math startup, launches an AI chatbot app

Maxwell Zeff — 3:27 PM PDT · July 28, 2025

Harmonic, an AI startup co-founded by Robinhood CEO Vlad Tenev, announced Monday the beta launch of an iOS and Android chatbot app where users can access its AI model, Aristotle.

With this launch, the company aims to broaden access to Aristotle, which Harmonic claims to offer “hallucination-free” answers for questions involving mathematical reasoning — a bold claim given the reliability problems of today’s AI models. Harmonic is focused on creating “mathematical superintelligence” or MSI; the startup eventually wants to help users with all fields that rely on math, including physics, statistics, and computer science.

“[Aristotle] is the first product available to people that does reasoning and formally verifies the

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output,” said Harmonic CEO and co-founder Tudor Achim in an interview with TechCrunch. “Within the domains that Aristotle supports, which are quantitative reasoning domains, we actually do guarantee that there’s no hallucinations.”

Eventually, Harmonic says it also plans to release an API to let enterprises access Aristotle, as well as a web app for consumers.

Harmonic says Aristotle achieved gold medal performance on the 2025 International Math Olympiad (IMO) through a formal test (meaning the problems were translated into a machine-readable format). Google and OpenAI also [developed AI models that achieved gold medal performance on this year’s IMO](#), but through informal tests taken in natural language.

Harmonic says it’s not releasing other benchmarks for Aristotle at this time.

The beta launch of Aristotle comes just a few weeks since Harmonic raised \$100 million in a Series B round led by Kleiner Perkins at an \$875 million valuation. Achim claims that Harmonic is “advancing very rapidly along” its path to achieving MSI and that investors believed that was a fair valuation given the scope of his startup’s ambition.

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Several leading tech companies are focused on training their AI models to solve math problems. AI that can do math is valuable in its own right, but math is also considered a uniquely verifiable domain that demands core reasoning skills. Systems that develop these capabilities may prove useful in other domains as well.

Achim says that Harmonic achieves its hyper-accurate solutions by having Aristotle produce responses in the open source programming language Lean. Before Aristotle gives an answer to users, he says the model double-checks that the solution is correct through an algorithmic process that doesn't involve AI. Harmonic's CEO notes that similar technology is used to verify outputs in high-stakes fields such as medical devices and aviation.

Even in a narrow domain, achieving hallucination-free performance from an AI model is an incredibly difficult task. Studies have found that [even leading AI models hallucinate a lot](#), and the problem doesn't appear to be getting better. [OpenAI's latest AI reasoning models hallucinate more than its older ones](#).

Topics: [AI](#) [AI chatbot](#) [AI math](#)





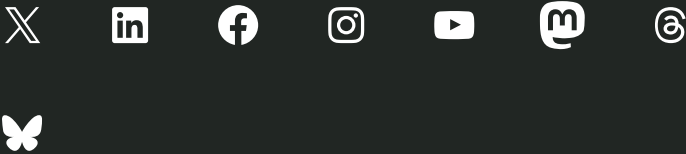
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